

Problems and conjectures

SGA 2, Exposé XIII

English draft

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About this draft

This is an unofficial English translation of Exposé XIII, “Problems and conjectures”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ XIII

Problems and conjectures

1. Relations between global and local results. Affine problems related to duality

It is well known that many statements concerning a projective scheme X can be formulated in terms of statements concerning a certain graded ring, or better a complete local ring, namely the homogeneous coordinate ring of X (i.e. the affine ring of the projecting cone $\tilde{X}$ of X), or its completion (i.e. the completion of the local ring of the vertex of $\tilde{X}$). The interest of this reformulation is that it often allows one, starting from known global results, to conjecture, or even to prove, analogous results for complete noetherian local rings more general than those which actually intervene in the global statement, for example for local rings which are not necessarily of equal characteristic. Thus, Serre's duality theorem for projective space XII XII.1.1 suggested the useful local duality theorem V V.2.1. Serre's fundamental theorem on the cohomology of coherent Modules on projective space (finiteness, asymptotic behaviour for large n of $H^i(X, F(n))$, cf. EGA III 2.2.1) generalizes to a structure theorem for the local invariants $H_m^i(M)$, see V V.3. Likewise, the Lefschetz theorems for the fundamental group, and the Picard group ("equivalence criteria"), quite familiar in the classical case and subsequently extended to an arbitrary base field, suggested the "local" Lefschetz theorems of exposés X and XI. Of course, the local theorems in their turn are valuable tools for obtaining global statements. For example local duality allows one to formulate a global asymptotic property XII XII.1.3 (i) by the vanishing of certain local invariants $H^i(F_x)$. In a more substantial way, the local Lefschetz theorems, implying for example the "purity" or the parafactoriality of certain complete intersection local rings (X X.3.4 and XI XI.3.13) allow one in the global Lefschetz theorems to do away with certain non-singularity hypotheses, as in X X.3.5, X.3.6, X.3.7.

Another useful generalization of the theorems concerning projective schemes over a field k consists in replacing k by a general base scheme. Thus, the sequel to EGA III will give¹ a generalization in this sense of Serre duality²; the finiteness theorems and the asymptotic behaviour of the $H^i(X, F(n))$ have been stated in EGA III 2.2.1 over a general base scheme, finally the Lefschetz theorems can likewise be developed for a projective morphism, as one has seen in XII XII.4.9, thanks to the local theorem XII XII.4.7. Of course, the fact of working over a general base scheme also leads to essentially new statements, such as the "comparison theorem" EGA III 4.15 and the existence theorem for sheaves EGA III 5.1.4 (which, as one has seen elsewhere in IX, arise from the same key theorems of cohomological nature as the Lefschetz theorems for π_1 and Pic).

It is then necessary to extract theorems which encompass simultaneously the two generalizations we have just indicated of the statements concerning projective schemes over a field. The natural objects for such a common generalization are the *noetherian rings separated and complete for an I -adic topology*. Their study, from this point of view, has not yet been seriously undertaken, and seems to me at the present time the most interesting subject in the local theory of coherent sheaves. Here is a typical problem in this direction:

¹**N.D.E.** in fact, this generalization is not to be found there; see the note below, and the editor's note (noteconrad) on page noteconrad

²cf. Hartshorne's seminar, cited at the end of Exp. IV.

²This conjecture, and conjecture 1.2 below, are false, as shown by R. Hartshorne, "Affine duality and cofinite modules", *Invent. Math.* **9** (1969/70), p. 145-164, section 3.

CONJECTURE 1.1. [*“Second affine finiteness theorem”^(**)3*] Let M be a module of finite type over a noetherian ring A (which if need be one will assume to be a quotient of a regular ring), J an ideal of A , prove that the modules $H_J^i(M)$ are “ J -cofinite”, i.e. that the modules

$$\mathrm{Hom}_A(A/J, H_J^i(M))$$

are of finite type.

Recall that one denotes by $H_J^i(M)$ the module $H_Y^i(X, \widetilde{M})$ (where $X = \mathrm{Spec}(A)$, $Y = V(J)$) of Exp. I, interpreted in II in terms of an inductive limit of cohomologies of Koszul complexes, or again for $i \geq 2$ the module $H^{i-1}(X - Y, \widetilde{M})$. Actually, 1.1 should be a consequence of a more precise statement, implying that the $H_J^i(M)$ lie in a suitable abelian subcategory $\underline{D}_J$ of the category $\underline{C}_J$ of A -modules of support $\subset Y = V(J)$, such that $H \in \mathrm{Ob} \underline{D}_J$ implies that H is J -cofinite. (N.B. The category of modules H with support contained in $V(J)$ and which are J -cofinite is unfortunately not stable under passage to quotients!). The essential problem would then consist in defining $\underline{D}_J$. More precisely, the solution of the problem 1.1 should emerge (at least if A is a quotient of a regular ring) from a duality theory, generalizing at once local duality, and the duality theory of projective morphisms alluded to above, and which would be of the following kind:

CONJECTURE 1.2. [*“Affine duality”⁴*] Suppose A regular, separated and complete for the J -adic topology. Let $C^\bullet(A)$ be an injective resolution of A .

(i) Prove that the functor

$$D_J : L_\bullet \longmapsto \mathrm{Hom}_J(L_\bullet, C^\bullet(A))$$

from the category of complexes of A -modules, free of finite type in every dimension and with degrees bounded above, (where the morphisms are the homomorphisms of complexes up to homotopy) to the category of complexes of A -modules $K^\bullet$, injective in every dimension and with degrees bounded above (where the homomorphisms are defined in the same way), is fully faithful.

(ii) Prove that for every $K^\bullet$ of the form $D_J(L_\bullet)$, the $H^i(K^\bullet) (= \mathrm{Ext}_Y^i(X; L_\bullet, \mathcal{O}_X))$ are J -cofinite.

(iii) More precisely, prove that the $K^\bullet$ which are homotopic to a complex of the form $D_J(L_\bullet)$ can be characterized by finiteness properties of the $H^i(K^\bullet)$, stronger than that envisaged in (ii), for example by the property $H^i(K^\bullet) \in \mathrm{Ob} \underline{D}_J$, where $\underline{D}_J$ is a suitable abelian category, as envisaged above.

Let us note that the problem is solved in the affirmative when A is local and J is an ideal of definition of A (cf. Exp. IV), and also when J is the zero ideal. In these two cases, exceptionally, one may restrict oneself to taking for $\underline{D}_J$ the category of Modules with support $V(J)$ which are J -cofinite, (which in the second case simply means that one takes the category

³**N.D.E.** however, if A is complete local (resp. regular of positive characteristic) and J is the maximal ideal, the statement is true for M of finite type (resp. $M = A$), cf. (Hartshorne R., “Affine duality and cofinite modules”, *Invent. Math.* **9** (1969/70), p. 145-164, corollary 1.4) (resp. (Huneke C. & Sharp R., “Bass numbers of local cohomology modules”, *Trans. Amer. Math. Soc.* **339** (1993), no. 2, p. 765-779), which moreover contains much stronger results). For completely different methods (D -modules) allowing one to approach characteristic zero, see (Lyubeznik G., “Finiteness properties of local cohomology modules (an application of D -modules to commutative algebra)”, *Invent. Math.* **113** (1993), no. 1, p. 41-55); see also by the same author “Finiteness properties of local cohomology modules for regular local rings of mixed characteristic: the unramified case”, *Comm. Algebra* **28** (2000), no. 12, p. 5867-5882, Special issue in honor of Robin Hartshorne, and “Finiteness properties of local cohomology modules: a characteristic-free approach”, *J. Pure Appl. Algebra* **151** (2000), no. 1, p. 43-50. The notion of cofinite module has since evolved under Hartshorne’s guidance. One says that M is J -cofinite if its support is contained in $V(J)$ and if all the $\mathrm{Ext}_A^i(A/J, H_J^i(M))$ are of finite type. On this subject, see for example (Delfino D. & Marley Th., “Cofinite modules and local cohomology”, *J. Pure Appl. Algebra* **121** (1997), no. 1, p. 45-52).

⁴This conjecture, false as it stands, has however been established in a fairly similar form by R. Hartshorne, “Affine duality and cofinite modules”, *Invent. Math.* **9** (1969/70), p. 145-164.

of Modules of finite type over A). An affirmative solution of conjecture 1.2 in general would give one for 1.1, by taking for $L_\bullet$ the dual of a free resolution of finite type of M . On the other hand, an affirmative solution of 1.1 would give an affirmative answer to the first part of the following conjecture which we formulate in “global” form:

CONJECTURE 1.3. *Let $X \subset \mathbf{P}_k^r$ be a closed subscheme of the standard projective scheme which is locally a complete intersection and every irreducible component of which is of codimension $\geq s$. Let $U = \mathbf{P}_k^r - X$.*

(i) *Prove that for every coherent Module F on U , one has*

$$\dim H^i(U, F) < +\infty \text{ for } i \geq s.^5$$

(ii) *Give an example, with X connected and regular, where one has*

$$H^s(U, F) \neq 0.$$

To see that (i) is a special case of 1.1 one considers

$$H^i(U, F(\cdot)) = \bigoplus_n H^i(U, F(n)) = H^i(E^{r+1} - \tilde{X}, \tilde{F})$$

as a module over the affine ring $k[t_0, \dots, t_r]$ of the projecting cone E^{r+1} of $\mathbf{P}^r$. This module is none other than $H_J^{i+1}(M)$, where J is the ideal of the projecting cone $\tilde{X}$ of X in E^{r+1} . On the other hand from the hypothesis made on X , which implies that $\tilde{X}$ is also a complete intersection of codimension $\geq s$ at every point of E^{r+1} distinct from the origin, it follows that $H_J^{i+1}(M)$ vanishes outside the origin for $i \geq s$. If therefore it is J -cofinite as 1.1 would have it, it is *a fortiori* $\mathfrak{m}$ -cofinite, which easily implies that it is finite-dimensional in every degree⁶. Note moreover that conjecture 1.3 already arises for a non-singular irreducible curve X in $\mathbf{P}^3$, one does not know whether in this case the $H^2(\mathbf{P}^3 - X, \mathcal{O}_X(n))$ are finite-dimensional, or whether they are necessarily zero⁷. One does not even know whether there exists an irreducible curve of $\mathbf{P}^3$ which is not set-theoretically the intersection of two hypersurfaces⁸.

PROBLEM 1.4. *Give an affine variant of the “comparison theorem” EGA III 4.1.5 as a theorem of commutation of the functors H_j^i with certain projective limits.*

Finally, in the present order of ideas, I had posed the following problem: let A be a regular complete noetherian local ring, K its field of fractions, prove that $\text{Ext}_A^i(K, A) = 0$ for every i . An affirmative answer was given on the spot by M. Auslander, the regularity hypothesis may be replaced by the hypothesis that A is integral and in fact it is true that $\text{Ext}_A^i(K, M) = 0$ for every i , as soon as M is of finite type over A . This follows at once from the following statement, due to Auslander: if A is a complete noetherian local ring, then for

⁵Part (i) of this conjecture is proved by R. Hartshorne when X is smooth cf. *Ample subvarieties of algebraic varieties*, Notes written in collaboration with C. Musili, Lect. Notes in Math, vol. 156, Springer-Verlag, Berlin-New York, 1970, theorem III.5.2. This author has also found an example for (ii), cf. R. Hartshorne, “Cohomological dimension of algebraic varieties”, *Ann. of Math. (2)* **88** (1968), p. 403–450, example page 449.

⁶**N.D.E.** Hartshorne has proved (Hartshorne R., “Cohomological dimension of algebraic varieties”, *Ann. of Math. (2)* **88** (1968), p. 403–450) that the cohomology $H^{n-1}(\mathbf{P}_k^n - X, F)$ vanishes for F coherent and X of positive dimension (k algebraically closed). In fact, thanks essentially to Serre duality and Lichtenbaum’s theorem — vanishing of the cohomology of coherent sheaves in maximal dimension for irreducible non-complete quasi-projective varieties —, one is reduced to proving that the formal completion $\hat{\mathbf{P}}_k^n$ and X have the same field of rational functions. This is the difficult point (theorem 7.2 of *loc. cit.*), in other words $\mathbf{P}_k^n$ is G_3 in the terminology of (Hironaka H. & Matsumura H., “Formal functions and formal embeddings”, *J. Math. Soc. Japan* **20** (1968), p. 52–82). These authors have independently proved the preceding results, and in fact much better. They have proved that X is universally G_3 and have computed the field of rational functions of the formal completion of an abelian variety along a subvariety of positive dimension. It is in this article that there appear for the first time the conditions G_1, G_2, G_3 now classical.

⁷The question has just been solved in the affirmative by R. Hartshorne (Hartshorne R., “Ample vector bundles”, *Publ. Math. Inst. Hautes Études Sci.* **29** (1966), p. 63–94, theorem 8.1) and H. Hironaka.

⁸**N.D.E.** see conjecture XII.3.5 and the corresponding note.

every module M of finite type over A , the functors $\text{Ext}_A^i(., M)$ transform inductive limits into projective limits⁹.

2. Problems related to π_0 : local Bertini theorems

Let A be a complete noetherian local ring, f an element of its maximal ideal, $X = \text{Spec}(A)$, $Y = \text{Spec}(A/fA)$. The use of the local “Lefschetz” technique allows one to give criteria for $Y' = X' \cap Y$ (where $X' = X - \{\mathfrak{m}\}$) to be connected, in terms of hypotheses on X' . Thus, it suffices that one have: a) X' connected, b) $\text{prof } \mathcal{O}_{X',x} \geq 2$ for every closed point x of X' , c) f is A -regular. One will note however that hypotheses b) and c) are not of a purely topological nature, for example are not invariant upon replacing X by X_{red} . In the analogous situation for a projective scheme X' over a field and a hyperplane section Y' of X' , the use of the “Bertini theorem” and of Zariski’s “connectedness theorem” allows one in fact to obtain results of a distinctly more satisfactory appearance, which had led me in the oral seminar to state a conjecture, which I have since resolved in the affirmative. Let us therefore state here:

THEOREM 2.1. *Let A be a complete noetherian local ring, X its spectrum, $\mathfrak{a}$ the closed point of X , $X' = X - \{\mathfrak{a}\}$. Suppose that X satisfies the conditions (where k denotes an integer ≥ 1):*

- a_k) *The irreducible components of X' are of dimension $\geq k + 1$.*
- b_k) *X' is connected in dimension $\geq k$, i.e. one cannot disconnect X' by a closed subset of dimension $< k$ (cf. III III.3.8).*

Let m be an integer, $0 \leq m \leq k$, and let $f_1, \dots, f_m \in \mathfrak{r}(A)$, set $B = A/\sum_i f_i A$, $Y = \text{Spec}(B) = V(f_1) \cap \dots \cap V(f_m)$, $Y' = X' \cap Y = Y - \{\mathfrak{a}\}$. Then Y satisfies the conditions a_{k-m} , b_{k-m} . In particular, for every sequence of $m \leq k$ elements $f_1, \dots, f_m$ of $\mathfrak{r}(A)$, $Y' = X' \cap V(f_1) \cap \dots \cap V(f_m)$ is connected.

It is moreover easy to see that if the last conclusion is valid (it evidently suffices to take $m = k$ therein), and excluding the case where X would be irreducible of dimension 0 or 1, it follows that the irreducible components of X' are of dimension $\geq k + 1$, and X' is connected in dimension $\geq k$, so that in a sense, 2.1 is a “best possible” result.

Let us give the principle of the proof of 2.1. Only the condition b_{k-m} for Y presents a problem. For given k one is easily reduced to the case where X is integral, and even (passing to the normalization, which is finite over X) to the case where X is *normal*. If $k = 1$, hence $\dim X' \geq 2$, then X' has depth ≥ 2 at its closed points and one may apply the result recalled at the beginning of the no., which shows that $Y' = X' \cap V(f)$ is connected. In the case $k \geq 1$, one assumes the theorem proved for the $k' < k$. By induction on m , one is reduced to the case where $m = 1$, i.e. to verifying that for $f_1 \in \mathfrak{r}(A)$, $X' \cap V(f_1)$ is connected in dimension $\geq k - 1$. If it were not, i.e. if it were disconnected by a Z' of dimension $< k - 1$, there would exist a sequence $f_2, \dots, f_k$ such that $X' \cap V(f_1) \cap \dots \cap V(f_k)$ is disconnected, and in this sequence one may choose $f_2 \in \mathfrak{r}(A)$ arbitrarily subject only to the condition of not vanishing at any point of a certain finite subset F of X' (namely the set of maximal points of Z'). Moreover, one verifies easily, using the fact that X' is normal, hence satisfies Serre’s condition (S_2) ¹⁰ that there exists a finite subset F' of X' such that $f \in \mathfrak{r}(A)$, $V(f) \cap F' = \emptyset$ implies that $V(f) \cap X'$ also satisfies the condition (S_2) . One can then choose f_2 so that f_2 vanishes neither on F nor on F' , hence so that $X' \cap V(f_2)$ satisfies (S_2) . But then, by virtue of Hartshorne’s theorem III III.3.6, $X' \cap V(f_2)$ is connected in codimension 1 hence (as every component of $X' \cap V(f_2)$ is of dimension $\geq k$) it is connected in dimension $\geq k - 1$. Applying the induction hypothesis to $V(f_2) = \text{Spec}(A/f_2 A)$, it follows that $X' \cap V(f_2) \cap V(f_1) \cap V(f_3) \cap \dots \cap V(f_k)$ is connected, whereas one had constructed it disconnected, absurd.

Let us mention some interesting corollaries:

⁹**N.D.E.** write $K = \varinjlim_{a \neq 0} A[1/a]$ and observe that $a \neq 0$ is A -regular.

¹⁰Cf. EGA IV 5.7.2.

COROLLARY 2.2. *Let $f: X \rightarrow Y$ be a proper morphism of locally noetherian preschemes, with Y integral, $y_0 \in Y$, y_1 the generic point of Y . One assumes*

- a) *Y is unibranch at y_0 , every irreducible component of X dominates Y .*
- b) *The irreducible components of X_1 are of dimension $\geq k + 1$, and X_1 is connected in dimension $\geq k$.*

Then the irreducible components of X_0 are of dimension $\geq k + 1$, and X_0 is connected in dimension $\geq k$.

Indeed, Zariski's connectedness theorem (cf. EGA III 4.3.1) implies that X_0 is connected; to show that it is not disconnected by a closed subset of dimension $< k$, one is reduced to showing that these local rings at points $x \in X_0$ such that $\dim \bar{x} < k$ have a spectrum not disconnected by x . Now this is true without assuming either that f is proper, or that Y is unibranch at y_0 . To see this one is reduced to the case where X is integral dominating Y , and if one wishes, Y affine of finite type over $\mathbf{Z}$, so that one is under the conditions of the dimension formula for $\mathcal{O}_{X,x}$ over $\mathcal{O}_{Y,y_0}$. Using in this case the finiteness of the normal closure, one may even assume X normal, hence by virtue of a theorem of Nagata¹¹, the completion of a local ring $\mathcal{O}_{X,x}$ of X' is still normal, hence (if $\mathcal{O}_{X,x}$ is of dimension N) $\text{Spec}(\widehat{\mathcal{O}}_{X,x})$ is connected in dimension $\geq N - 1$. Let $n = \dim \mathcal{O}_{Y,y_0}$, then $\deg \text{tr } k(x)/k(y) < k$ implies $\dim \mathcal{O}_{X,x} > n + (k + 1) - k = n + 1$, taking into account $\dim X_1 \geq k + 1$, and taking a system $f_1, \dots, f_n$ of parameters of $\mathcal{O}_{Y,y_0}$ which one lifts to elements of $\mathcal{O}_{X,x}$, one sees by 2.1 that $\text{Spec}(\widehat{\mathcal{O}}_{X,x} / \sum f_i \widehat{\mathcal{O}}_{X,x})$ is connected in dimension ≥ 1 , i.e. is not disconnected by its closed point, or what amounts to the same, $\text{Spec}(\widehat{\mathcal{O}}_{X_0,x})$ is not disconnected by its closed point; *a fortiori* the same holds for $\text{Spec}(\mathcal{O}_{X_0,x})$.

As in the case of the ordinary connectedness theorem, one can vary 2.2 by taking the geometric fibres (over the algebraic closures of the residue fields), provided one assumes Y geometrically unibranch at y_0 , or (with no hypothesis other than that Y is noetherian) that f is universally open. Applying this to the case where Y is the dual scheme of a projective scheme $\mathbf{P}_k^r$ over a field, one recovers a strengthened form of the global result which had inspired 2.1, namely:

COROLLARY 2.3¹². *Let X be a closed subscheme of $\mathbf{P}_k^r$ (k a field), one assumes the irreducible components of X are of dimension $\geq l + 1$, and X geometrically connected in dimension $\geq l$. Then for every sequence $H_1, \dots, H_m$ of m hyperplanes of $\mathbf{P}_k^r$ ($0 \leq m \leq l - 1$), $X \cap H_1 \cap \dots \cap H_m$ satisfies the same condition with $l - m$, in particular is geometrically connected in dimension $\geq l - 1$.*

One can moreover modify this statement in an evident way, for the case where one is given a proper morphism $X \rightarrow \mathbf{P}_k^r$, which is not necessarily an immersion, and an analogous extension is possible for 2.1 (by considering a scheme proper over X'). These statements are moreover formally deduced from the statements given here, taking into account the ordinary connectedness theorem which reduces us to the case of a finite morphism.

COROLLARY 2.4. *Let A be a complete normal noetherian local ring of dimension $\geq k + 2$. Let $X = \text{Spec}(A)$, $X' = X - \{\mathfrak{a}\}$, $f_1, \dots, f_k$ be elements of $\mathfrak{r}(A)$, then $Y' = X' \cap V(f_1) \cap \dots \cap V(f_k)$ is connected, and $\pi_1(Y') \rightarrow \pi_1(X')$ is surjective.*

One proceeds as in SGA 1 X 2.11.

In all of this, it was only a matter of questions of *connectedness*. Now in the global case, well-known theorems assert that for an irreducible projective variety $X \subset \mathbf{P}_k^r$, k algebraically closed, its intersection with a sufficiently "general" hyperplane H is irreducible, (and not

¹¹Cf. EGA IV 7.8.3 (i) (ii) (v).

¹²**N.D.E.** for a very beautiful direct proof, see (Fulton W. & Lazarsfeld R., "Connectivity and its applications in algebraic geometry", in *Algebraic geometry (Chicago, Ill., 1980)*, Lect. Notes in Math., vol. 862, Springer, Berlin-New York, 1981, p. 26–92, theorem 2.1). Cf. also [HL], cited in the editor's note (46) page 15.

merely connected): this is the *Bertini theorem*, proved by Zariski, which implies in turn by Zariski's connectedness theorem, that for *every* H , $H \cap X$ is connected (although not necessarily irreducible). One can moreover proceed in the opposite direction, by proving this last result by a Lefschetz-type technique, and by deducing the Bertini theorem, reducing to the case where X is normal, and using the following result: for H “sufficiently general”, $X \cap H$ is likewise normal. This suggests the

CONJECTURE 2.5¹³. *Let A be a complete normal noetherian local ring. Show that there exists a non-zero $f \in \mathfrak{r}(A)$ such that $Y' = X' \cap V(f) = Y - \{\mathfrak{a}\}$ (where $Y = \text{Spec}(A/fA)$) is normal (hence irreducible by 2.1 if $\dim A \geq 3$).*

To do things properly, one should show that in a suitable sense, there even exist “many” elements f having the property in question, for example that one can choose f in an arbitrary power of the maximal ideal. Using Serre's criterion for normality, and the remark made above for Serre's property (S_2) , one sees that one would have an affirmative answer to 2.5 if one had one to the

CONJECTURE 2.6¹⁴. *Let A be a complete noetherian local ring, U an open subset of its spectrum X , F a finite subset of $X' = X - \{\mathfrak{a}\}$. One assumes that U is regular. Prove that there exists an $f \in \mathfrak{r}(A)$ such that $V(f) \cap U$ is regular, and $V(f) \cap F = \emptyset$.*

See, for a result of “local Bertini” type, Chow [2]

3. Problems related to π_1

Here again, one has numerous questions, suggested by the global results or the transcendental results.

CONJECTURE 3.1¹⁵. *Let A be a complete noetherian local ring with algebraically closed residue field, $X = \text{Spec}(A)$, $X' = X - \{\mathfrak{a}\}$, $\mathfrak{a}$ the closed point. Suppose the irreducible components of X are of dimension ≥ 2 , X' connected.*

- (i) *Prove that $\pi_1(X')$ is topologically finitely generated.*
- (ii) *If p is the characteristic exponent of the residue field k of A , prove that the largest topological quotient group of $\pi_1(X')$ which is “of order prime to p ” is of finite presentation.*

For part (i), using descent theory SGA 1 IX 5.2 and theorem 2.4, one is reduced to the case where A is normal of dimension 2. In this case, a systematic method for studying the fundamental group of X' , inaugurated by Mumford [5] in the transcendental setting, consists in *desingularizing* X i.e. in considering a projective birational morphism $Z \rightarrow X$, with Z integral regular, inducing an isomorphism $Z' = Z|X' \rightarrow X'$; it is plausible that such a Z always exists, this is in any case what is shown by Abhyankar's method [1] in the case of “equal characteristics”¹⁶. Let C be the fibre of the closed point of X under $Z \rightarrow X$, it is an algebraic curve over the residue field k , connected by virtue of the connectedness theorem. The solution of 3.1 then seems tied to the

¹³N.D.E. see the following editor's note.

¹⁴N.D.E. one now finds a proof of this conjecture in the literature, and hence the preceding one should likewise be regarded as proved as indicated above. One can also find two attempts at proofs, published earlier but alas unsuccessful, by Flenner and Trivedi. See Trivedi V., “Erratum: “A local Bertini theorem in mixed characteristic””, *Comm. Algebra* **25** (1997), no. 5, p. 1685-1686. However, the editor has not verified that the proof is now complete.

¹⁵N.D.E. the analogous statement is true for (connected) schemes X of finite type over a separably closed field k under the hypothesis of strong desingularization for all k -schemes (of finite type), in particular if k is of characteristic zero or X of dimension ≤ 2 . To this end one is reduced to the case of quasi-projective surfaces by Lefschetz-type techniques developed by Mme Raynaud, cf. notes *supra*; see SGA 7.1, theorem II.2.3.1.

¹⁶The possibility of “resolving” A is now proved in complete generality by Abhyankar [8].

PROBLEM 3.2. *With the preceding notations, relate $\pi_1(X')$ with the topological invariants of C , in particular its fundamental group, (so as to bring out the topological finite generation of $\pi_1(X')$, using for example SGA I, theorem X 2.6).*

Another method would be to consider A as a finite algebra over a complete *regular* local ring B of dimension 2, ramified along a curve C contained in $\mathrm{Spec}(B) = Y$. One is thus led to the

PROBLEM 3.3. *Let A be a complete regular local ring of dimension 2, with algebraically closed residue field k , X its spectrum, C a closed subset of X of dimension 1. Define local invariants of the immersed curve C , having a meaning independent of the residual characteristic, knowledge of which allows one to compute the fundamental group of $X - C$ by generators and relations when k is of characteristic zero. Prove that when k is of characteristic $p > 0$, the “tame” fundamental group of $X - C$ is a quotient of the former, and the two fundamental groups (in characteristic 0, and in characteristic $p > 0$) have the same maximal quotient of order prime to p .*

Of course, 3.3 shows us that in 3.1, one should likewise replace X' by a scheme of the form $X - Y$, where Y is a closed subset of X which is of codimension ≥ 2 in every component of X containing it. When one drops this restriction on Y , there should still exist an analogous finiteness result, provided one puts restrictions of “tame” type on the ramification at the maximal points of those irreducible components of Y which are of codimension 1.

PROBLEM 3.4. *Let A be a complete noetherian local ring of dimension 2, with algebraically closed residue field. Let again $X = \mathrm{Spec}(A)$, $X' = X - \{\mathfrak{a}\}$. Find particular structural properties of $\pi_1(X')$ for the case where A is a complete intersection.*

A satisfactory solution of this problem might perhaps allow one to solve the following old problem:

CONJECTURE 3.5¹⁷. *Find an irreducible curve in $\mathbf{P}_k^3$ (k an algebraically closed field), preferably non-singular, which is not set-theoretically the intersection of two hypersurfaces.*

(Kneser [4] shows that one can always obtain it as the intersection of three hypersurfaces).

4. Problems related to the higher π_i : local and global Lefschetz theorems for complex analytic spaces¹⁸

Let X be a scheme locally of finite type over the field of complex numbers $\mathbf{C}$, one can associate to it an analytic space X^h over $\mathbf{C}$, whence homotopy and homology invariants $\pi_i(X^h)$, $H_i(X^h)$, $H^i(X^h)$ etc. Moreover one knows that X is connected if and only if X^h is, hence that one has a bijection

$$\pi_0(X^h) \longrightarrow \pi_0(X)$$

Likewise, since every étale covering X' of X defines an étale covering X'^h of X^h , one has a canonical homomorphism

$$\pi_1(X^h) \longrightarrow \pi_1(X),$$

of which one knows, using a theorem of Grauert–Remmert, that it identifies the second group with the compactification of the first for the topology of subgroups of finite index (which simply expresses the fact that $X' \mapsto X'^h$ is an equivalence of the category of étale coverings of X with the category of *finite* étale coverings of X^h). It follows that the results of this seminar (by a purely algebraic route) on $\pi_0(X)$ and $\pi_1(X)$ imply results for $\pi_0(X^h)$ and $\pi_1(X^h)$ (which are of a transcendental nature). If moreover X is proper, the well-known exact sequence $0 \rightarrow \mathbf{Z} \rightarrow \mathbf{C} \rightarrow \mathbf{C}^* \rightarrow 0$ allows one to show that the Néron–Severi group of X

¹⁷**N.D.E.** this problem is, as of autumn 2004, still open.

¹⁸**N.D.E.** the statements are made more precise in the Comments (section 6). The conjectures which appear there have become theorems, cf. the footnotes of section 6.

(the quotient of its Picard group by the connected component of the identity) is isomorphic to a subgroup of $H^2(X^h, \mathbf{Z})$; in the nonsingular Kähler case, this is the subgroup denoted $H^{(1,1)}(X^h, \mathbf{Z})$ (classes of type $(1, 1)$):

$$\text{Pic}(X)/\text{Pic}^0(X) \subset H^2(X, \mathbf{Z}).$$

Consequently, the information we have obtained on Picard groups implies information, very partial it is true, on the groups $H^2(X^h, \mathbf{Z})$. It is tempting to complete all these fragmentary results by conjectures.

Very precise indications, going in the same direction as those just mentioned, are furnished by a classical Lefschetz theorem [7]. It asserts that if X is a nonsingular irreducible projective analytic space of dimension n , and if Y is a nonsingular hyperplane section, then the injection

$$Y^{n-1} \longrightarrow X^n$$

induces a homomorphism

$$\pi_i(Y^{n-1}) \longrightarrow \pi_i(X^n)$$

which is an *isomorphism* for $i \leq n - 2$, an *epimorphism* for $i = n - 1$. There follows the analogous statement for the homomorphisms

$$H_i(Y^{n-1}) \longrightarrow H_i(X^n)$$

on homology (integral, to fix ideas), while in cohomology,

$$H^i(X^n) \longrightarrow H^i(Y^{n-1})$$

is an isomorphism in dimension $i \leq n - 2$, a monomorphism in dimension $i = n - 1$. We have obtained variants of these results in the setting of schemes, for π_0 , π_1 , Pic , valid moreover without non-singularity hypotheses to a large extent, cf. Exp. XII. Moreover, in the elimination of the non-singularity hypotheses, we have used in an essential way “local” variants of these global Lefschetz theorems. All this suggests the following problems, which will doubtless have to be attacked simultaneously¹⁹.

PROBLEM 4.1. *Let X be an analytic space, Y a closed analytic subset of X (or simply a closed subset?)²⁰ such that for every $x \in Y$, the local ring $\mathcal{O}_{X,x}$ is a complete intersection. Let n be the complex codimension of Y in X . Is the canonical homomorphism*

$$\pi_i(X - Y) \longrightarrow \pi_i(X)$$

an isomorphism for $i \leq n - 2$, and an epimorphism for $i = n - 1$?

In this problem and the following ones, one of course implicitly assumes a base-point chosen in order to define the homotopy groups. To state the following problem, one must define, for an analytic space X (more generally, for a locally path-connected space) and an $x \in X$, local invariants $\pi_i^x(X)$ ²¹. For this, one chooses a non-constant map f from the interval $[0, 1]$ into X , such that $f(0) = x$ and $f(t) \neq x$ for $t \neq 0$ (such exist if x is not an isolated point). Then for every neighbourhood U of x , there exists an $\varepsilon > 0$ such that $0 < t < \varepsilon$ implies $f(t) \in U$, and the homotopy groups $\pi_i(U - x, f(t))$ are essentially independent of t (they are, as t varies, related by a transitive system of isomorphisms), one may denote them $\pi_i(U - x, f)$. One then sets

$$\pi_i^x(X) = \varprojlim_U \pi_{i-1}(U - x, f)$$

the projective limit being taken over the system of open neighbourhoods U of x . Strictly speaking, this limit depends on f , and should be denoted $\pi_i^x(X, f)$, but one verifies that

¹⁹The formulations 4.1 to 4.3 which follow are provisional. See conjectures A to D below, in “comments on Exp. XIII” for more satisfactory formulations, as well as Exp. XIV.

²⁰**N.D.E.** the meaning of this question is not clear; indeed, the very statement of the problem does not seem to make sense in this case since the codimension of Y in X is not defined when Y is no longer assumed analytic.

²¹If $i \geq 2$. For the case $i \leq 1$, cf. *Comments* in no. 6 below, page 14.

as f varies, these groups are isomorphic to one another²², more precisely they form a local system on the space of paths of the type envisaged issuing from x . These invariants are the homotopical version of the local cohomology invariants $H_x^i(F)$ for a sheaf F on X , introduced in I, and should play the role of *relative local homotopy groups* of X modulo $X - x$. Their vanishing for $i \leq n$ and for every $x \in Y$, where Y is a closed subset of X of topological dimension $\leq d$, should imply that the homomorphisms

$$\pi_i(X - Y) \longrightarrow \pi_i(X)$$

are bijective for $i < n - d$, and surjective for $i = n - d$ ²³. From this point of view, 4.1 would imply (for Y reduced to a point) a conjecture of a purely local nature, expressed by

$$\pi_i^x(X) = 0 \text{ for } i \leq n - 1$$

when X is a complete intersection of dimension n at x .

As an example of local invariants $\pi_i^x(X)$, let us note that if x is a nonsingular point of X of complex dimension n , then

$$\pi_i^x(X) = \pi_{i-1}(S^{2n-1}),$$

where S^{2n-1} denotes the sphere of dimension $2n - 1$. In particular in this case $\pi_i^x(X) = 0$ for $i \leq 2n - 1$, which corresponds to the fact that if from a topological *manifold* X one removes a closed subset Y of codimension $\geq m$, then $\pi_i(X - Y) \rightarrow \pi_i(X)$ is an isomorphism for $i \leq m - 2$ and an epimorphism for $i = m - 1$.

This being said:

PROBLEM 4.2. *Let X be an analytic space, $x \in X$, t a section of $\mathcal{O}_X$ vanishing at x , Y the set of zeros of t . Suppose the following conditions are satisfied:*

- a) *t is regular at x (i.e. not a zero-divisor at x , a hypothesis perhaps superfluous, moreover).*
- b) *At the points x' of $X - Y$ neighbouring x , $\mathcal{O}_{X,x'}$ is a complete intersection (a hypothesis which should be replaceable by the following more general one if 4.1 is true: for x' as above, $\pi_i^{x'}(X) = 0$ for $i \leq n - 1$).*
- c) *At the points y of $Y - \{x\}$ neighbouring x , one has*

$$\text{prof } \mathcal{O}_{X,y} \geq n$$

(it suffices for example that one have $\text{prof } \mathcal{O}_{X,x} \geq n$).

Under these conditions, is the canonical homomorphism

$$\pi_i^x(Y) \longrightarrow \pi_i^x(X)$$

an isomorphism for $i \leq n - 2$, an epimorphism for $i = n - 1$?

Here finally is a global variant of 4.2, which should be deduced from it by consideration of the projecting cone at its vertex, and which would generalize the classical Lefschetz theorems:

PROBLEM 4.3. *Let X be a projective analytic space, equipped with an ample invertible Module L , t a section of L , Y the set of zeros of t . Suppose:*

- a) *t is a regular section (a hypothesis perhaps superfluous).*
- b) *For every $x \in X - Y$, $\mathcal{O}_{X,x}$ is a complete intersection (should be replaceable by $\pi_i^x(X) = 0$ for $i \leq n - 1$).*
- c) *For every $x \in Y$, $\text{prof } \mathcal{O}_{X,x} \geq n$.*

Under these conditions, is the homomorphism

$$\pi_i(Y) \longrightarrow \pi_i(X)$$

an isomorphism for $i \leq n - 2$, an epimorphism for $i = n - 1$?

²²At least if x does not disconnect X in a neighbourhood of x , cf. *Comments* below, page 14.

²³For a corrected formulation, cf. *Comments* below, page 14.

We leave to the reader the task of stating analogous conjectures of a *cohomological* nature²⁴, the hypotheses and conclusions then bearing on the local cohomological invariants (with coefficients in a given group). In any case, the key result seems as though it should be 4.2, when hypothesis *b*) is taken there in the respected form, — whether one places oneself at the point of view of homology, or of homotopy.

We have stated these conjectures in the transcendental setting, in the hope of interesting topologists in them and of convincing them that questions of “Lefschetz” type are far from being closed. Of course, now that we are on the point of having at our disposal a good theory of the cohomology of schemes (with finite coefficients), thanks to the recent work of M. Artin, the same questions arise in the setting of schemes, and it is hard to doubt that they will receive a positive answer, in the near future²⁵.

5. Problems related to local Picard groups

A first fundamental problem, pointed out for the first time by Mumford [5] in a particular case, is the following. Let A be a complete local ring with residue field k , $X = \text{Spec}(A)$, $U = \text{Spec}(A) - \{a\}$, where a is the maximal ideal of A i.e. the closed point of $\text{Spec}(A)$. One proposes to construct a strict projective system G of locally algebraic groups G_i over k , and a natural isomorphism

$$(+) \quad \text{Pic}(U) \simeq G(k)$$

where one of course sets $G(k) = \varprojlim G_i(k)$. Heuristically, one proposes to “put an algebraic group structure” (or, at least, a pro-algebraic one, in a suitable sense) on the group $\text{Pic}(U)$.

It is clear that as it stands, the problem is not precise enough, since the datum of an isomorphism (+) is far from characterizing the pro-object G . If A contains a subfield still denoted k , which is a field of representatives, one can make the problem more precise, by requiring that for a variable extension k' of k , one have an isomorphism, functorial in k' :

$$(+') \quad \text{Pic}(U') \simeq G(k')$$

where U' is the open set analogous to U in $\text{Spec}(A')$, $A' = A \hat{\otimes}_k k'$. One can proceed in an analogous way even if A has no field of representatives, provided k is perfect, which then allows one to construct functorially an A' “by residual extension k'/k ”. Moreover, when A admits a field of representatives, the algebraic structure that one will find on $\text{Pic}(U)$ will depend essentially on the choice of this field of representatives (as one already sees on the projecting cone of an elliptic curve), it therefore seems that one must start from an “algebraic pro-ring over k ” in the manner of Greenberg [3], in order to arrive at a definition of the pro-object G . It is moreover conceivable that in the case where there is no given field of representatives, one finds only a projective system of *quasi*-algebraic groups in the sense of Serre, or rather quasi-locally algebraic groups (the groups G_i obtained will not in general be of finite type over k , but only locally of finite type over k). It is even possible that in general one will find only a still weaker structure on $\text{Pic}(U)$, of the kind encountered by Néron [6] in his theory of degeneration of abelian varieties defined over local fields.

A method for attacking the problem, also introduced by Mumford, consists in desingularizing X , i.e. in considering a projective birational morphism $Y \rightarrow X$ with Y regular. When U is regular (i.e. a is an isolated singular point), one can often find Y in such a way that $Y|_U = V \rightarrow U$ is an isomorphism. In this case, one will therefore have

$$\text{Pic}(U) \simeq \text{Pic}(V) \simeq \text{Pic}(Y) / \text{Im } \mathbf{Z}^I,$$

where I is the set of irreducible components of the fibre Y_a (each of these defining an element of $\text{Pic}(Y)$, being a locally principal divisor, thanks to Y being regular). On the other hand,

²⁴Cf. Exp. XIV the corresponding results in the theory of schemes.

²⁵see the previous footnote.

using the technique of Formal Geometry EGA III 4 and 5, notably the existence theorem, one finds

$$\mathrm{Pic}(Y) \simeq \varprojlim \mathrm{Pic}(Y_n),$$

where $Y_n = Y \otimes_A A_n$, $A_n = A/\mathfrak{m}^{n+1}$. When A admits a field of representatives k , one has at one's disposal the theory of Picard schemes of the projective schemes Y_n over k , hence one has

$$\mathrm{Pic}(Y_n) \simeq \mathbf{Pic}_{Y_n/k}(k).$$

This therefore furnishes a construction of a projective system of locally algebraic groups $\mathbf{Pic}_{Y_n/k}/\mathrm{Im} \mathbf{Z}^I$, which is the desired system.²⁶ In the case envisaged here, one can moreover see (using that a is an isolated singular point) that the connected components of the universal image subgroups in this projective system form an *essentially constant* projective system, so in this instance one finds a locally algebraic group G as solution of the problem. If one even assumes A normal of dimension 2, then a remark of Mumford (saying that the intersection matrix of the components of Y_a in X is negative definite²⁷) implies that G is even an algebraic group, i.e. of finite type over k (the number of its connected components being moreover equal to the determinant of the intersection matrix considered a moment ago).

If on the other hand a is not an isolated singularity, one convinces oneself on examples (with A of dimension 2) that one finds a projective system of algebraic groups, not reducing to a single algebraic group.

Once one had at one's disposal a good notion of “*local Picard scheme*”, there would be occasion to strengthen the notion of parafactoriality, by saying that A is “*geometrically parafactorial*”, when not only A and even $\hat{A}$ are parafactorial, but the local Picard scheme $G(\hat{A})$ is the trivial group (which is stronger, when the residue field is not algebraically closed, than saying that G has no rational point over k other than the identity). One realizes the necessity of a strengthened notion of parafactoriality, by recalling that there exist complete normal local rings of dimension 2 which are factorial, but which admit finite étale algebras which are not²⁸. A local ring “*geometrically factorial*” would then be a normal ring A such that all the localizations of dimension ≥ 2 are geometrically parafactorial, or better, such that the localizations of $\hat{A}$ are parafactorial²⁹. Of course, it would be interesting to find a “good” definition of these notions, independent of the theory, still to be developed, of local Picard schemes.³⁰

It is in any case plausible that one will need these notions if one wishes to obtain statements of the following type: Let A be a “good ring” (for example an algebra of finite

²⁶**N.D.E.** the question has been greatly clarified by the results of Boutot (Boutot J.-F., *Schéma de Picard local*, Lect. Notes in Math., vol. 632, Springer, Berlin, 1978). In particular, if A is a complete (noetherian) local k -algebra of depth ≥ 2 such that $H_{\mathfrak{m}}^2(A)$ is finite-dimensional over k , the local Picard group is a group scheme locally of finite type over k , with tangent space at the origin $H_{\mathfrak{m}}^2(A)$. If A is moreover normal of dimension ≥ 3 , Serre's criterion for normality XI.3.11 together with corollary V.3.6 ensure the required finiteness and, hence, the existence of the local Picard scheme. See also (Lipman J., “The Picard group of a scheme over an Artin ring”, *Publ. Math. Inst. Hautes Études Sci.* **46** (1976), p. 15–86) for an approach closer to that of Grothendieck sketched above.

²⁷**N.D.E.** Mumford D., “The topology of normal singularities of an algebraic surface and a criterion for simplicity”, *Publ. Math. Inst. Hautes Études Sci.* **9** (1961), p. 5–22.

²⁸**N.D.E.** factorial rings with non-factorial henselization arise naturally when one studies moduli spaces of vector bundles. See for example (Drézet J.-M., “Groupe de Picard des variétés de modules de faisceaux semi-stables sur $\mathbf{P}_2$ ”, in *Singularities, representation of algebras, and vector bundles* (Lambrecht, 1985), Lect. Notes in Math., vol. 1273, Springer, Berlin, 1987, p. 337–362). *Stricto sensu*, Drézet shows that the completion is not factorial, but in fact the proof gives the result for the henselization: the point is that Luna's étale slice theorem (Luna D., “Slices étales”, in *Sur les groupes algébriques*, Mém. Soc. math. France, vol. 33, Société mathématique de France, Paris, 1973, p. 81–105) describes the local ring of a quotient in the sense of invariant geometry near a semi-stable point locally for the étale topology.

²⁹For a more flexible notion of “geometrically factorial” local ring, cf. *Comments*, page 13.

³⁰**N.D.E.** see page 13: a local ring is geometrically factorial (resp. parafactorial) if its strict henselization is factorial (resp. parafactorial).

type over $\mathbf{Z}$, or over a complete local ring, for example over a field). Let U be the set of $x \in X = \text{Spec}(A)$ such that $\mathcal{O}_{X,x}$ is “geometrically factorial”, then U is open? Or again: Let $f: X \rightarrow Y$ be a flat morphism of finite type with Y locally noetherian, let U be the set of $x \in X$ such that $\mathcal{O}_{X_{f(x)},x}$ is “geometrically factorial”, then U is open, at least under pleasant additional conditions on f ? I doubt that with the usual notion of factorial ring, there exist true statements of this type.

We have raised here, in a particular case, the question of the study of the geometric properties of “variable” local rings, for example the $\mathcal{O}_{X,x}$ as x runs through a prescheme X . When X is a scheme of finite type over a field, for example, one knows³¹ that there exists on X a projective system of finite algebras $P^n_{X/k}$ (obtained by completing $X \times_k X$ along the diagonal), whose fibre at every point $x \in X$ rational over k is isomorphic to the projective system of the $\mathcal{O}_{X,x}/\mathfrak{m}_X^{n+1}$. It is then natural to relate the study of the completions of the local rings $\mathcal{O}_{X,x}$, as x varies, to that of the “algebraic family of complete local rings” given by the P^n , noting that for every $x \in X$ (rational over k or not), one obtains a complete local ring

$$P^\infty(x) = \varprojlim P^n(x)$$

(where $P^n(x) = \text{reduced fibre } P^n \otimes_{\mathcal{O}_{X,x}} k(x)$). Particular interest will attach for example to the complete ring thus associated to the generic point, and one will expect that its algebro-geometric properties (expressed for example by its local Picard, or homotopy, or homology, groups), will be essentially those of the completions $\hat{\mathcal{O}}_{X,x}$ for x in a suitable dense open set U .

One can, in a general way, propose to undertake the simultaneous study of the complete local rings thus obtained from an adic projective system (P_n) of finite algebras over a given scheme X . It is plausible that one will find, subject to certain regularity conditions (such as the flatness of the P_n), that the local homotopy groups come from a projective system of finite group schemes over X , and that one will have analogous results for the local Picard groups. As regards the latter, a first interesting case which deserves to be investigated is that where one starts from an algebraic surface X having singular curves, and one proposes to study the local Picard schemes at the variable points on these, in terms of a suitable group pro-scheme defined on the singular locus.

6. Comments³²

The point of view of “étale cohomology” of schemes and the recent progress in this theory lead us to make more precise and at the same time to broaden certain of the problems posed. For the notion of “topology” and of “étale topology of a scheme”, I refer to M. Artin, *Grothendieck Topologies*, Harvard University 1962 (mimeographed notes)³³.

This theory, by a finer notion of localization than that furnished by the traditional “Zariski topology”, leads one to attach particular interest to *strictly local* rings, i.e. henselian local rings with separably closed residue field. For every local ring A with residue field k , and every separable closure k' of k , one can find a local homomorphism of A into a strictly local ring A' , the *strictly local closure* of A , with residue field k' , having an obvious universal property. A' is henselian, flat over A , and $A' \otimes_A k \simeq k'$; it is noetherian if and only if A is. (Cf. *loc. cit.* Chap. III, section 4)³⁴. If X is a prescheme, and x a point of X , x' a point over x , spectrum of a separable closure k' of $k = k(x)$, one is led to define the *strictly local ring* of X at x' , $\mathcal{O}'_{X,x'}$, as the strictly local closure of the usual local ring $\mathcal{O}_{X,x}$, relative to the residual extension k'/k . It is the *strictly local* rings of the “geometric” points of X which, from the point of view of the étale topology, are supposed to reflect the local properties of the prescheme X . They also play, in many respects, the role that one used to have the *completions* of the local

³¹N.D.E. see EGA IV.16.

³²Written in March 1963.

³³Or preferably, to SGA 4.

³⁴Or EGA IV 18.8.

rings of X play (say, at the points with algebraically closed residue field), while remaining “closer” to X and allowing an easier passage to “neighbouring points”.

There is then occasion to take up again a good number of questions, which one generally poses for complete local rings (possibly restricted to having an algebraically closed residue field), for noetherian henselian local rings (resp. noetherian strictly local rings). Thus the topological problems raised in nos. 2 and 3 arise more generally for strictly local rings. One can moreover state conjecturally, for “good” strictly local rings, certain properties of simple connectedness and of acyclicity for the geometric fibres of the canonical morphism $\mathrm{Spec}(\hat{A}) \rightarrow \mathrm{Spec}(A)$, which would show that for many properties of a “topological” nature, it comes to the same to prove them for the ring A , or for its completion $\hat{A}$. Certain results already obtained in this direction³⁵ allow one to hope that one will soon have at one’s disposal complete results in this direction.

The notion of étale localization furnishes a definition which seems reasonable of the notion of “*geometrically parafactorial*” or “*geometrically factorial*” local ring (whose need was pointed out in no. 5, p. 11): one will call thus a local ring whose strictly local closure is parafactorial, resp. factorial. Hypotheses of this nature do in fact enter in a natural way in the study of the étale cohomology of preschemes³⁶. Thus, if X is a locally noetherian prescheme whose strictly local rings are factorial (i.e. whose ordinary local rings are “geometrically factorial”), one shows that the $H^i(X_{\text{ét}}, \mathbf{G}_m)$ are torsion groups for $i \geq 2$ (which sometimes allows one to express these groups in terms of the cohomology groups with coefficients in the groups μ_n of n -th roots of unity), and if X is integral with fraction field K , the natural homomorphism $H^2(X_{\text{ét}}, \mathbf{G}_m) \rightarrow H^2(K, \mathbf{G}_m) = \mathrm{Br}(K)$ is injective³⁷ examples show that these conclusions may fail, even for X local, if one assumes only that X is factorial instead of geometrically factorial³⁸.

Concerning the problems of local and global Lefschetz type raised in 3.4, and their analogues in the theory of schemes, the homological version of these questions has been considerably clarified, everything resulting formally from three general theorems, one concerning the cohomological dimension of certain affine schemes (resp. of Stein spaces), such as affine schemes X of finite type over an algebraically closed field: their cohomological dimension is $\leq \dim X$ (“affine Lefschetz theorem”)³⁹: the other being a duality theorem for the cohomology (with discrete coefficients) of a projective morphism⁴⁰, finally the last a theorem of *local duality* of an analogous nature⁴¹. In Algebraic Geometry, only this last

³⁵Cf. M. Artin in SGA 4 XIX

³⁶**N.D.E.** see for example (Strano R., “The Brauer group of a scheme”, *Ann. Mat. Pura Appl.* (4) **121** (1979), p. 157–169) where the hypothesis of geometric parafactoriality of the local rings of a scheme X sometimes allows one to show the coincidence of the Brauer groups of X (computed in terms of Azumaya algebras) and the cohomological Brauer group of X .

³⁷**N.D.E.** the link between Brauer group and Picard group is intimate. Let us cite in this connection the following results of Saito (Saito S., “Arithmetic on two-dimensional local rings”, *Invent. Math.* **85** (1986), no. 2, p. 379–414) in the case of surfaces, the first being local the other global. Let A be an excellent local ring of dimension 2, normal and henselian with *finite* residue field and X the complement of the closed point in $\mathrm{Spec}(A)$. Then one has a perfect duality of *torsion* groups $\mathrm{Pic}(X) \times \mathrm{Br}(X) \rightarrow \mathbf{Q}/\mathbf{Z}$ — by Brauer group of X , one means the *cohomological* Brauer group $\mathrm{Br}(X) = H_{\text{ét}}^2(X, \mathbf{G}_m)$. In the global case, one has the following generalization of a result of Lichtenbaum (Lichtenbaum S., “Duality theorems for curves over p -adic fields”, *Invent. Math.* **7** (1969), p. 120–136): let k be the fraction field of a complete discrete valuation ring $\mathcal{O}$ with finite residue field and X a projective, smooth and geometrically complete curve over k . The group $\mathrm{Pic}^0(X)$ is equipped with the topology induced from the adic topology of k and $\mathrm{Pic}(X)$ is the topological group which makes $\mathrm{Pic}^0(X)$ an open subgroup. Then one has a perfect duality of topological groups $\mathrm{Pic}(X) \times \mathrm{Br}(X) \rightarrow \mathbf{Q}/\mathbf{Z}$. Note that this statement, which concerns curves, is of course proved by considering a regular (proper and flat) model of X over $\mathcal{O}$: it is a result on surfaces.

³⁸Cf. A. Grothendieck, the Brauer group II (Séminaire Bourbaki no. 297, Nov. 1965), notably 1.8 and 1.11 b.

³⁹Cf. SGA 4 XIV.

⁴⁰Cf. SGA 4 XVIII.

⁴¹Cf. SGA 5 I.

is not proved at the moment of writing these lines (it is however in characteristic 0, using the resolution of singularities by Hironaka). Moreover, in the transcendental setting, one has at present at one's disposal the global and local duality, proved recently by Verdier⁴². Let us confine ourselves to indicating that in the statement of the homological versions of problems 4.2 and 4.3 (which now deserve the name of conjectures), the conditions “at infinity” a) and c) are certainly superfluous, only the *local cohomological structure* of $X - Y$ being important, which one will assume for example locally a complete intersection of dimension $\geq n$. Moreover, in 4.3 say, the fact that Y is a hyperplane section should not play a role, and should be replaceable by the sole hypothesis that X is compact and $X - Y$ is Stein (i.e. in the case of Algebraic Geometry, X is proper over k and $X - Y$ affine; as we were saying, the homological version of this conjecture is proved for algebraic spaces over the field $\mathbf{C}$)⁴³.

In the definition (p. 8) of the $\pi_i^x(X)$, one must assume $i \geq 2$. For $i = 0, 1$, there is no reasonable definition of the $\pi_i^x(X)$; there is occasion to replace them by $H_0^x(X)$ and $H_1^x(X)$, defined respectively as the cokernel and the kernel in the natural homomorphism

$$\varprojlim H_0(U - (x), \mathbf{Z}) \longrightarrow \varprojlim H_0(U, \mathbf{Z}).$$

At a pinch and for the convenience of the formulations, one may set $\pi_i^x(X) = H_i^x(X)$ for $i \leq 1$, otherwise one must complete the later assertions concerning the π_i^x by the corresponding assertions for H_0^x, H_1^x . If x is an isolated point of X , it is appropriate to set $\pi_i^x(X) = 0$ for $i \neq 0$, $\pi_0^x(X) = H_0^x(X) = \mathbf{Z}$.

The assertion that the $\pi_i^x(U, f)$ are isomorphic to one another is true only when X is not disconnected by x in a neighbourhood of x , i.e. if $\pi_i^x(X) = 0$ for $i = 0, 1$. In the general case $\pi_i^x(X)$ can denote only a *family* of groups, not necessarily isomorphic to one another; however the writing $\pi_i^x(X) = 0$ retains an obvious meaning.

Page 9, where I foresee that the vanishing of the local homotopical invariants $\pi_i^x(X)$ for $x \in Y$, $i \leq n$ should imply the bijectivity of $\pi_i(X - Y) \rightarrow \pi_i(X)$ for $i < n - d$, the surjectivity for $i = n - d$, it is appropriate to be cautious, for want of being able to have at one's disposal in the present context (as in Algebraic Geometry) “generic” points at which the local conditions will also have to apply. It will doubtless be necessary, for this reason, to appeal to relative local homotopical invariants

$$\pi_i^Y(X, f) = \pi_i^Y(X, x) = \varprojlim_U \pi_{i-1}(U - U \cap Y, f(t)) \quad \text{for } i \geq 2,$$

(and an *ad hoc* definition as above for $i = 0, 1$), where Y is a closed subset of X ; or to make up for the absence of generic points by expressing the hypotheses on X in terms of properties of a topological nature (for the étale topology) of the spectra of the local rings of X , which allows one to recover generic points. The same reservation applies to the generalization of conjectures 4.2 and 4.3 to the case where $X - Y$ is not assumed locally a complete intersection, a generalization suggested in the statement of conditions b) of these conjectures.

To formulate the expurgated versions of conjectures 4.2 and 4.3 suggested by the results alluded to above, it is appropriate to set down the

DEFINITION 1. Let X be a topological space, Y a locally closed subset of X , and n an integer. One says that X is of *homotopical depth* $\geq n$ along Y , and one writes $\text{prof hpt}_Y(X) \geq n$, if for every $x \in Y$, one has $\pi_i^Y(X, x) = 0$ for $i < n$.

It should be equivalent to say that for every open set X' of X , and every $x \in X' \cap U = U'$ (where $U = X - Y$), the canonical homomorphism

$$\pi_i(U', x) \longrightarrow \pi_i(X', x)$$

⁴²**N.D.E.** see Verdier J.-L., “Dualité dans la cohomologie des espaces localement compacts”, in *Séminaire Bourbaki*, vol. 9, Société mathématique de France, Paris, 1995, Exp. 300, p. 337–349.

⁴³Cf. Exp. XIV

is an isomorphism for $i < n - 1$, a monomorphism for $i = n - 1$ ⁴⁴.

DEFINITION 2. Let X be a complex analytic space, n an integer, one says that the *rectified homotopical depth* of X is n ⁴⁵, if for every locally closed analytic subset Y of X , one has

$$(x) \quad \text{prof htp}_Y(X) \geq n - \dim Y$$

(where, of course, $\dim Y$ denotes the *complex* dimension of Y).

It should be equivalent to say that for every irreducible analytic subset Y locally closed in X , there exists a closed analytic subset Z of Y , of dimension $< \dim Y$, such that the relation (x) is valid for $Y - Z$ in place of Y . This would allow one for example in definition 2 to confine oneself to the case where Y is nonsingular.⁴⁶

The following conjecture, of a purely topological nature, is in the nature of a “*local Hurewicz theorem*”.

CONJECTURE A. [“*local Hurewicz theorem*”⁴⁷] Let X be a topological space, Y a locally closed subset, subjected if need be to “smoothness” conditions of the kind of local triangulability of the pair (X, Y) , n an integer ≥ 3 . In order that one have $\text{prof htp}_Y(X) \geq n$, it is necessary and sufficient that one have

$$H_Y^i(\mathbf{Z}_X) = 0 \quad \text{for } i < n$$

(one then says that X is of cohomological depth $\geq n$ along Y), and that the local fundamental groups

$$\pi_2^Y(X, x) = \varprojlim_{U \ni x} \pi_1(U - U \cap Y)$$

be trivial (one then says that X is “pure” along Y).

One will note that if X is an analytic space, Y an analytic subspace, and if X is pure along Y , then for every $x \in Y$, the local ring $\mathcal{O}_{X,x}$, as well as its localizations with respect to prime ideals containing the ideal defining the germ Y at x (i.e. in the inverse image Y_x of Y under $\text{Spec}(\mathcal{O}_{X,x}) = X_x \rightarrow X$) are pure in the sense of Exp. X; it seems plausible that the converse is equally true. Analogous remarks hold for cohomological depth, it being understood that one works with the étale topology on the $\text{Spec}(\mathcal{O}_{X,x})$.

Conjecture 4.1 then generalizes to

CONJECTURE B. [“*Purity*”⁴⁸] Let E be an analytic space, X an analytic subset of E . One assumes that E is nonsingular of dimension N at $x \in X$, and that X can be described by p analytic equations in a neighbourhood of every point. Then the rectified homotopical depth of X is $\geq N - p$.

In particular, a local complete intersection of dimension n at every point would be of rectified homotopical depth $\geq n$, which is nothing other than conjecture 4.1.

Conjectures 4.2 and 4.3 generalize respectively to:

⁴⁴**N.D.E.** when the pair (X, Y) is moreover polyhedral, this equivalence is true; cf. (Eyrat C., “Profondeur homotopique et conjecture de Grothendieck”, *Ann. Sci. Éc. Norm. Sup. (4)* **33** (2000), no. 6, p. 823–836).

⁴⁵In the first edition of these notes, we had used the term: “true homotopical depth”. In the present version, we follow EGA IV 10.8.1.

⁴⁶**N.D.E.** all the conjectures which follow, suitably rectified if I may say so, have become theorems thanks to the work of Hamm and Lê Dũng Tráng (Hamm H.A. & Lê Dũng Tráng, “Rectified homotopical depth and Grothendieck conjectures”, in *The Grothendieck Festschrift, Vol. II*, Progr. Math., vol. 87, Birkhäuser, Boston, 1990, p. 311–351) cited [HL] in what follows. As regards the two conjecturally equivalent definitions of the rectified depth, they are even equivalent to a third, which is expressed in terms of a Whitney stratification (cf. *loc. cit.*, theorem 1.4).

⁴⁷**N.D.E.** as observed in [HL], example 3.1.3, this conjecture is already false for $X = \{z \in \mathbf{C}^n \mid z_1^2 + z_2^2 + \dots + z_n^2 = 0\}$, $n \geq 4$ and Y reduced to the origin. But, suitably modified, it is true (theorem 3.1.4 of *loc. cit.*).

⁴⁸**N.D.E.** this conjecture is proved, even in the case where E is singular, in [HL]: it is theorem 3.2.1.

CONJECTURE C. [“local Lefschetz”⁴⁹] Let X be an analytic space, Y a closed analytic subset, x a point of Y , one assumes that $X - Y$ is Stein in a neighbourhood of x (for example Y defined by an equation at x), and that $X - Y$ is of rectified homotopical depth $\geq n$ in a neighbourhood of x (for example, is at every point of $X - Y$ neighbouring x a complete intersection of dimension $\geq n$, cf. conjecture B). Then the canonical homomorphism

$$\pi_i^x(Y) \longrightarrow \pi_i^x(X)$$

is an isomorphism for $i < n - 1$, an epimorphism for $i = n - 1$.

CONJECTURE D. [“global Lefschetz”⁵⁰] Let X be a compact analytic space, Y an analytic subspace of X such that $U = X - Y$ is Stein, and is of rectified homotopical depth $\geq n$ (for example a complete intersection of dimension $\geq n$ at every point). Then the canonical homomorphism

$$\pi_i(Y) \longrightarrow \pi_i(X)$$

is an isomorphism for $i < n - 1$, an epimorphism for $i = n - 1$.

REMARK. When, in the statements C and D, one replaces the hypothesis that $X - Y$ is Stein by the hypothesis that $X - Y$ is a union of $c + 1$ Stein open sets (which will play the role of a hypothesis of topological “concavity”), the conclusions should be modified simply by replacing n there by $n - c$.⁵¹

Let us finally spell out, in the “global case” D, the conjecture concerning the fundamental group (obtained by setting $n = 3$):

CONJECTURE D'. [global Lefschetz for the fundamental group⁵²] Let X be a compact analytic space over the field of complex numbers, Y a closed analytic subset such that $U = X - Y$ is Stein. Suppose moreover the following conditions are satisfied:

- (i) For every $x \in U$, the local fundamental group $\pi_2^x(X, x)$ is trivial (i.e. X is “pure at x ”), or merely the local ring $\mathcal{O}_{X,x}$ is pure.
- (ii) The local rings of the points of U are “connected in dimension ≥ 2 ”.
- (iii) The local rings of the points of U are of dimension ≥ 3 .

Under these conditions, for every $x \in Y$, the homomorphism

$$\pi_1(Y, x) \longrightarrow \pi_1(X, x)$$

is an isomorphism (and $\pi_2(Y, x) \rightarrow \pi_2(X, x)$ an epimorphism).

One will note that the local conditions (i) (ii) (iii) on U are satisfied if U is locally a complete intersection of dimension ≥ 3 . From the point of view of Algebraic Geometry, (when U comes from a scheme, still denoted U), conditions (i) to (iii) correspond to hypotheses on the local invariants $\pi_i^x(U)$, namely $\pi_i^x(U) = 0$ for $i < 3 - \deg \operatorname{tr} k(x)/k$, for the points x such that one has respectively $\deg \operatorname{tr} k(x)/k = 0, 1, 2$. The global condition on U (U Stein) will be satisfied if X is projective and Y a hyperplane section.

⁴⁹**N.D.E.** this conjecture is proved in [HL], this even in its strong form of the remark which follows, cf. theorem 3.3.1 of *loc. cit.*

⁵⁰**N.D.E.** this conjecture is here again proved in [HL], this even in its strong form of the remark which follows, cf. theorem 3.4.1 of *loc. cit.*

⁵¹**N.D.E.** let us finally mention the following result of Fulton, to be compared with the Fulton–Hansen result cited in the editor’s note (FH) page FH: let X and H be closed subschemes of $\mathbf{P}_{\mathbf{C}}^m$, n the dimension of X and d the codimension of H . Then the map

$$\pi_i(X, X \cap H) \longrightarrow \pi_i(\mathbf{P}_{\mathbf{C}}^n, H)$$

is an isomorphism if $i \leq n - d$ and is surjective if $i = n - d - 1$; see (Fulton W., “Connectivity and its applications in algebraic geometry”, in *Algebraic geometry (Chicago, Ill., 1980)*, Lect. Notes in Math., vol. 862, Springer, Berlin-New York, 1981, p. 26–92).

⁵²**N.D.E.** this conjecture is proved in [HL], cf. theorem 3.5.1 of *loc. cit.*

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